

Capturing project areas in Google Earth

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What?

Using the free software Google Earth to show your CCRP project area and research sites.

Why?

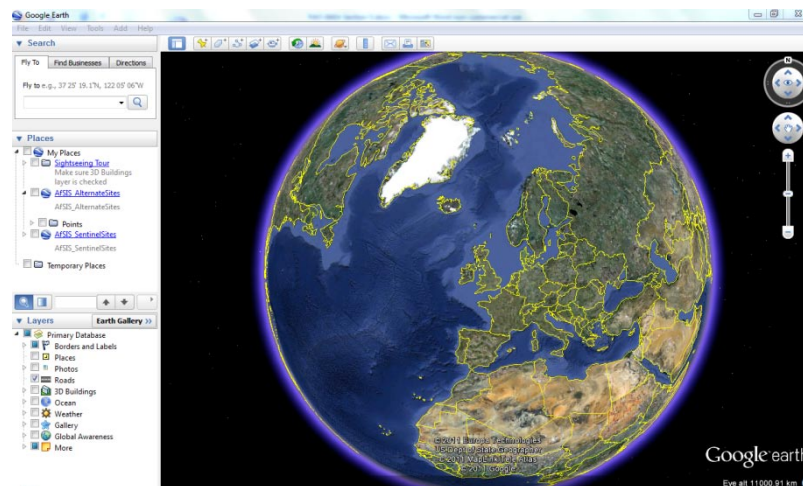
Three reasons:

1. You can use it to describe where you work, to include in reports and presentations and to learn about the area.
2. We will compile the results for all projects in the region to produce a regional picture of where CCRP is working.
3. You can transfer the information to other GIS systems and start to do more detailed mapping and characterization of the project – details on how to do that are coming soon.

How?

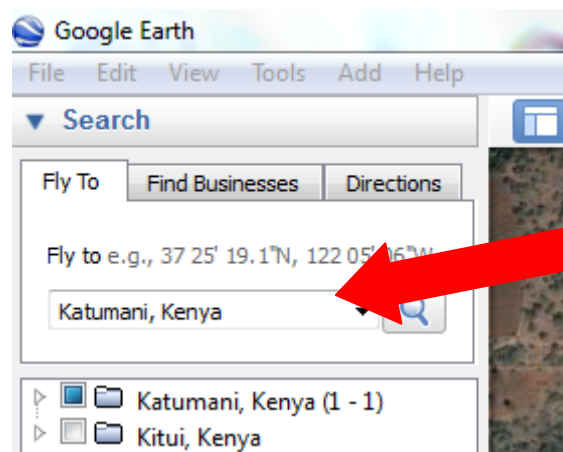
1. Use a computer with a reliable internet connection. It does not need to be really fast, but it does need to be working.
2. If Google Earth is installed go to Step 3. Otherwise:
 - a. Go to <http://www.google.com/earth/index.html>
 - b. Click on Download Google Earth 6 and follow the instructions for installing the software. It requires about 600Kb to be downloaded.
 - c. Browse the other information on the download page while you are there, so you can find out more about Google Earth.
3. Start Google Earth. You will get a screen that looks like Figure 1.

Figure 1



Try out the controls in the top right of the screen (arrowed in Figure 1) to move around and zoom in and out.

Figure 2

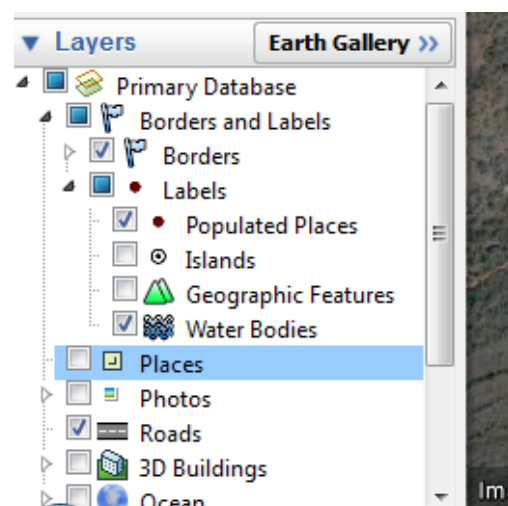


4. Try to find where you are working. You might want to search for a place name. Do that by typing the name of the place in the Fly To box (arrowed in Figure 2) and pressing Enter.

5. Google Earth does not know every place name, but it knows many. Try entering them as Place, Country. Now scroll around and try to identify the places you know. Note that Google Earth will need some time to refresh the screen when you scroll or zoom. It is downloading information from the internet, so the time it takes depends on the speed of your internet connection.

6. Make it easier to find your way by turning on or off relevant detail. In the Layers panel in the bottom left of the screen (Figure 3) you can select such things as roads, place names and administrative boundaries to be hidden or shown. As you find your way around and start to recognize the places you are working, see what you can learn about the landscape from the aerial viewpoint. Often it is

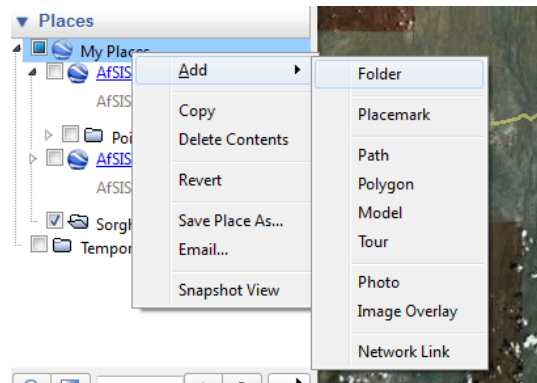
Figure 3



possible to see things such as patterns in land use that are not very obvious from the ground .

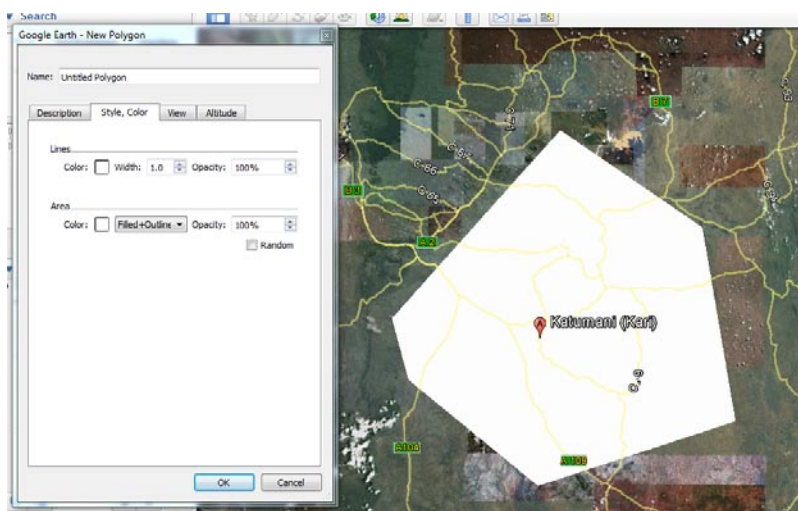
7. Now you are ready to capture information about your project area.
 - a. Zoom and scroll until you can see the project area on the screen.
 - b. You need to create folders

Figure 4



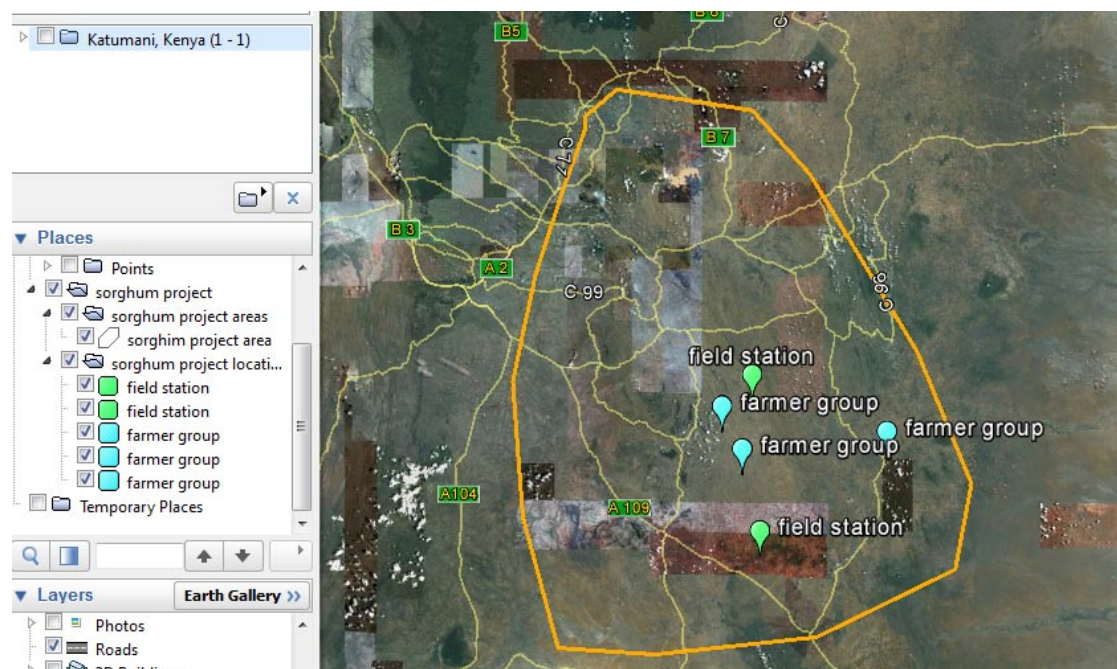
- Give it a name – for example Sorghum Project.
 - c. Now you need to make 2 subfolders inside that one. One will hold information about project areas – ie the boundaries of the region where you work, or that you hope your results will be relevant to. The other will hold information about specific locations – field stations or location of farmer groups, for example. Create these by right-clicking on the folder you created and select Add>Folder. Give them suitable names.
 - d. Right click on the folder you have just created for the project area and select Add>Polygon. Give it a name, such as Sorghum Project Area **but do not click OK yet!**
 - e. Put the cursor somewhere over the map window and it will appear as a small box with crosshairs. Put it at one corner of your project area and click. Now move it to another corner and click. Keep going around the boundary of your project area (Figure 5). A polygon marking your area will appear. Don't worry if it is not right first time. It is easy to delete and start again. It does not seem to be easy to edit a shape you have already drawn.

Figure 5



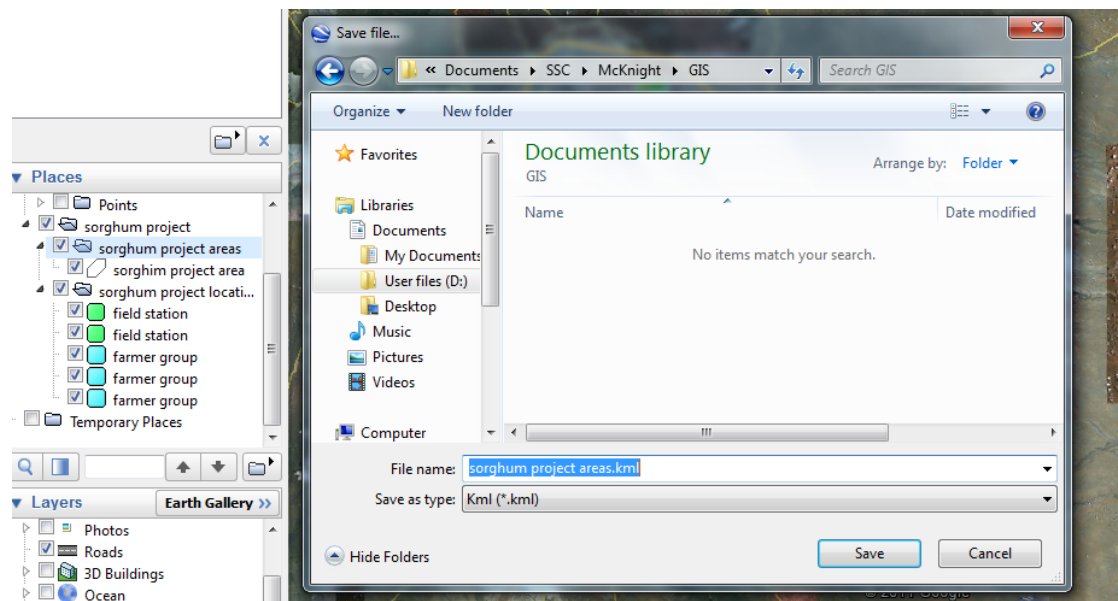
- f. Click on the New Polygon box for your area again. Select the Style tab and change Area to Outlined, so only the edge of your area is drawn and you can see the map inside. Try changing the colour and thickness too. Click OK.
- g. If your project area is made up of several disconnected patches, simply repeat from Step 7d for each patch.
- h. You can also add place markers to show the location of research sites, farmer groups or other important project locations. Right click on the second folder you made (Step 7c) and select Add>Placemaker. Click on the right place in the map, add a name, change the style or the icon (the symbol to the right of the name), and click OK. Repeat as often as you like. If something comes in the wrong place just delete (right click...Delete) and do it again. Eventually you might have something like Figure 6, in which I have the Sorghum project area marked with an orange line with the location of two field stations and three farmer groups plotted.

Figure 6



8. Now you are ready to export this information. In the Places panel, right click on the folder that contains the area information you added ('sorghum project areas' in Figure 6). Choose Save Place as... and select a suitable file name and location. Make sure the filetype is .kml (Figure 7). Files of type .kmz cannot be used in some GIS software. Repeat for the location folder ('sorghum project location' in Figure 6)

Figure 7



9. Last step. Send the two .kml files you have created by email to your regional representative (Theo or Linnet) so that the regional picture can be compiled. We should be able to send a compiled version back to you so that you can see where all the projects in the region are working.
10. But that was really just the beginning. Soon we will be sending out a starter pack that will allow you to do much more simple GIS work, including mapping your project area, characterizing it and describing results spatially. In the mean time, try to find what else you can use Google Earth for in your project and tell the rest of COP about it.

If you have a questions about this process try sending an email to Ric:
r.coe@cgiar.org